Arthur Galiullin

Solidity / Protocol Engineer — Warsaw, Poland (Remote)
work@arthurka.eu | github.com/arthurka-o | linkedin.com/in/arthur-galiullin | arthurka.eu
CET/CEST (UTC+1/+2) · Not relocating · Authorised to work in Poland · Open to B2B or employment

Solidity engineer with 4+ years of hands-on contract development and 8 years in production engineering. Shipped a perps DEX from testnet to TGE as one of six core engineers, took a Liquity fork multi-asset and upgradeable via UUPS, and built the referral, leaderboard and funded-trader risk systems around the protocol. Twice a hackathon prize winner. Strongest at smart contracts and protocol design.

WORK EXPERIENCE

Independent — Personal Projects *(June 2026 – Present)*

Working independently on personal projects.

Protocol Engineer — Overlay Protocol *(May 2024 – May 2026)*

Exotic perps DEX on BSC ($2.9M raised, backed by Polychain Capital, 1kx, ParaFi; audited by Spearbit, Trail of Bits, Nethermind). One of 6 core engineers through testnet to TGE.

- **Shiva Router Contract** — Contributed to the proxy router in front of Overlay's immutable core contracts. It is also where the protocol states which systems may call which functions, which is what lets the core stay immutable. Verified on-chain, 30,000+ transactions processed.
- **Immutable Price Oracle** — Rebuilt the feed on PancakeSwap V3 pools and removed its upgradeability in two steps until fully immutable, while every other contract stays upgradeable. A price bug sizes every loan wrong the moment it lands; resolution takes the worst of TWAP and spot, so manipulation can only ever hurt whoever performs it.
- **Security Testing** — Built the Foundry invariant suite (a trader's staked balance must never exceed the notional they can claim), fuzzed with Chimera, plus mainnet-fork tests for the oracle path. Ported the Chainlink feed tests from Python into Foundry, covering the inputs that broke things: a zero price, and a macro window equal to the micro window.
- **Funded Trader System** — Sponsored users trade with protocol-provided USDT, so the protocol is the risk manager. Safe multisigs with Zodiac modules keep funds to Overlay contracts and the allowed functions; off-chain monitoring closes every position and pauses the account on either of two thresholds — value below 50% of initial collateral, or a 25% drop since the day's first check. Split across two processes so the public API holds read-only database credentials.
- **On-Chain Referral System** — Integrated an already-audited referral contract and built the verification layer around it: off-chain signed messages checked against on-chain state, with a fallback that rebuilds a missing record from chain, so a database cleanup can no longer permanently disqualify a real affiliate.
- **Leaderboard & Trading Competitions** — Built the seasonal leaderboard (Node.js, MongoDB) as a configurable rule engine, so launching a new competition is a configuration change rather than a redeploy. The collateral floor is what stops a dust position from topping a board on leverage alone.
- **Subgraph & SDK** — Most of the protocol's indexing is mine: rewrote the config so new factories and networks stopped needing YAML hand-edits, added the router and PowerCard entities, built the Matchstick suite over the handlers, and made the index multichain including Berachain. Also maintain `overlay-sdk`, where USDT's 6 decimals must be handled against the 18 most tokens assume.
- **Frontend** — Contributor to the web app: fixed a 256-bit on-chain price coerced with `Number()`, silently losing precision in the two figures traders trust — value locked and unrealised PnL. Built the prediction-market screens.

Stack: Solidity, Foundry, Safe, Zodiac, TypeScript, Node.js, MongoDB, Ethers.js, React, Next.js, The Graph

Solidity Developer — KUMO *(March 2022 – January 2024)*

Lending protocol secured by carbon credits. Selected for **ABN AMRO + Techstars Future of Finance Accelerator**; built and launched a pilot with ABN AMRO. One of 2 Solidity developers.

- Joined KUMO on a **Liquity fork** retargeted at carbon credits and threaded the asset parameter through the protocol to make it multi-asset: every function touching a position now takes the asset it works with, plus a factory that creates a stability pool per asset instead of hand-copying contracts.
- Split the core contract's oversized functions to get past the EVM's stack limit, and researched the **Diamond Pattern (EIP-2535)** as the fix, prototyping the split. The protocol shipped on **UUPS** instead — the upgradeable-proxy ecosystem around it was the more mature one at the time.
- Added a **per-asset mint cap** on stablecoin creation. Writing the headroom calculation surfaced a real bug: when the cap sat below the amount already minted, the unsigned subtraction wrapped around and reported enormous room instead of none.
- Integrated **Chainlink and Tellor** oracle feeds for multi-source price validation. Upstream contribution to the original protocol, reviewed and merged by their team.
- Co-built the **carbon-credit lending contract** for the pilot, and I took most of it: a five-state loan lifecycle the contract refuses to leave out of order, explicit allowlists so money only moves between counterparties the bank approved, and liquidation proceeds routed to a treasury.

Stack: Solidity, Hardhat, Chainlink, Tellor, OpenZeppelin, Diamond Pattern, UUPS, Ethers.js

---

Ruby on Rails Developer *(2016 – 2022)*

Flatstack / Galvanize (2020–2022) — Enterprise GRC platform used by the Canadian Government: public API, async pipelines, N+1 performance work, DataDog and Airbrake monitoring. **meyvn** (2018–2020) — HR platform and news scoring engine; led the migration from a monolithic Rails app to a React SPA with a Rails API backend. **Technokratos** (2016–2017) — Virtual goods marketplace and photo albums: REST APIs, PDF generation, OpenCV face detection.

HACKATHONS

StreamBeam — ETHGlobal Prague 2025 *(Solo)* — **Prize Winner** (LayerZero General Prize)

Cross-chain crypto donations for livestream creators, with real-time OBS alerts on stream. Built solo in 36 hours, then developed for another six weeks.

InSure — ETHWarsaw 2025 *(Team)* — **Prize Winner**

On-chain insurance with automated claims and verification. Solidity, Foundry.

SKILLS

Protocol: Solidity, Foundry, Hardhat, EVM, UUPS, Diamond Pattern, OpenZeppelin, Safe, Zodiac, Merkle Trees, TypeChain

Oracles & Risk: Chainlink, Tellor, TWAP, invariant testing, Chimera, liquidation, collateral risk parameters

Backend: TypeScript, Node.js, Ethers.js, Viem, Ruby on Rails, PostgreSQL, MongoDB, GraphQL, The Graph, React, Next.js

Testing: Foundry (unit, invariant, fork), Chai, Mocha, Jest, Vitest, Playwright, Supertest, RSpec

Infra: AWS (Lambda, ECS, Terraform), Docker, GitHub Actions, CI/CD, Sidekiq, Git

---

EDUCATION

M.Sc. Computer Software Engineering — KFU, Institute of Information Technology (2018–2020)

B.Sc. Applied Informatics — KFU, Institute of Information Technology (2014–2018)

LANGUAGES

Russian (Native) | English (Professional) | Polish (Intermediate)

Shiva Router — BscScan | v1-shiva | v1-subgraph | overlay-sdk | KUMO protocol | StreamBeam | InSure